

FIT 5215 Deep Learning

Week 3: Revision, MLP Vs Sliding Window, Invariances

Dr. Arghya Pal
(email: arghya.pal)

Dr. Sailaja Rajanala
(email: sailaja.rajanala)

Chief Examiner: Dr. Trung Le
(email: trunglm)





Dr. Arghya Pal
(email: arghya.pal)



Dr. Sailaja Rajanala
(email: sailaja.rajanala)

* add

[@monash.edu](mailto:***@monash.edu)



Dr. Trung Le
(email: trunglm)

Assessments:

1. **Two in-semester online tests:** 10% each
 - a. During week 7 and week 12
 - b. Approximately 1 hour long, online test
2. **Two assignments:** 20% (A1) and 20% (A2)
 - a. Due in week 7 and week 12
3. **One final exam (closed book):** 40%
 - a. 10 minutes reading
 - b. Duration: 2 hours

In-tutorial Kaggle

1. Kaggle competitions, work in groups (3-5 members, no solo)
2. *If you surpass our quite trivial baseline ("**beat the boss**"), you will win 1 star*
3. *If you surpass our second baseline ("**beat the second boss**"), you will win 1 more star*
4. *Tutorial topper gets 3 star, Cohort topper gets 4 star.*
5. **Stars you earn can be converted to bonus points in assignments**, with 2 stars equaling 1 point, up to a maximum of 10 bonus points
6. Groups are fixed after week 4

Communication

* waiting time 2sec - 3 days



**Colab notebook
access:**

<https://identity.monash.edu/>

Note

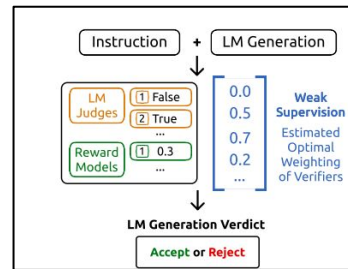
1. **Urgent emails** should enter with a clear title "FIT5215 [Urgent] [Your Name]. **Non-urgent email** should be returned back to you with 80 copies
2. We don't cheat during teaching, we expect same integrity from you!
3. All activities are offline, talk to SoIT office. We don't handle attendance related logistics like we don't handle Monash Building Constructions
4. If you get 79.8, it's 79.8, and not 80 !
5. We provide consultation only for FIT5215. Your MAI thesis supervisor should clear your doubt for your MAI minor thesis.

What is Supervised Learning?

Once upon a time, there was a little boy named Max. Max loved to play with his toy blocks. He had blocks of all colors and shapes. Max liked to build tall towers and big castles. One day, Max was playing with his blocks in his room. He built the tallest tower he had ever made. It was so tall that it almost touched the ceiling! Max was very proud of his tower. Suddenly,

Reference (GPT-4o): ... Suddenly, Max's cat, Whiskers, ran into the room. Whiskers saw the tall tower and thought it was a fun thing to play with. He jumped up and knocked the tower over. The blocks fell all over the floor. Max was very sad. He had worked so hard to build the tower, and now it was gone.

Fully Trained Transformer: ... Suddenly, Max's little sister came into the room. She wanted to play with Max's blocks too. But Max didn't want to share. He wanted to keep playing with his blocks. His sister started to cry and Max didn't understand why his sister was so upset. He thought he was just playing with them. So, he went to his room and looked for the blocks. After a while, he found them under the bed. Max was so happy that he hugged his sister and his sister. They played together and had a lot of fun. The end.



Difference, Distance, and Divergence

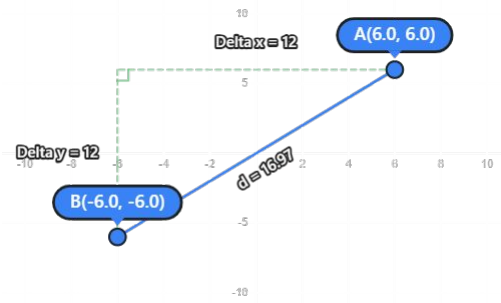
Difference

How much did the values change? Can be negative

$$MD = E[|X - Y|] = E_X[E_{Y|X}[|X - Y|]] = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n |x_i - y_j|$$

Distance

How far apart are two objects?
Generally non-negative

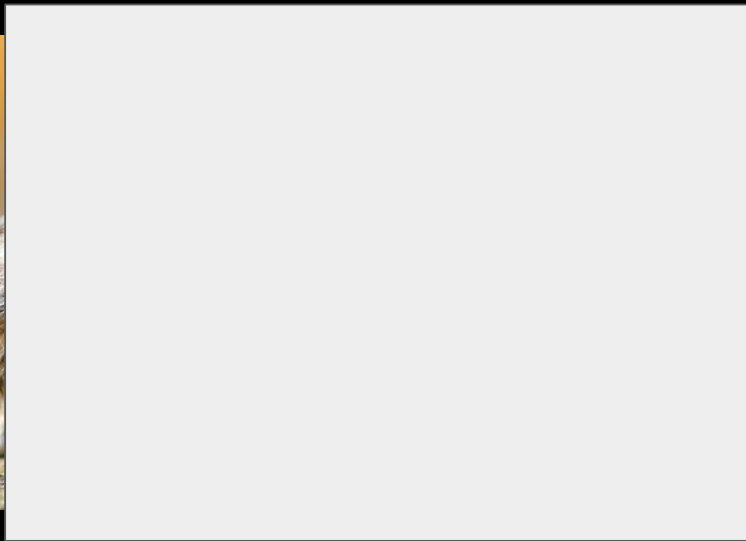
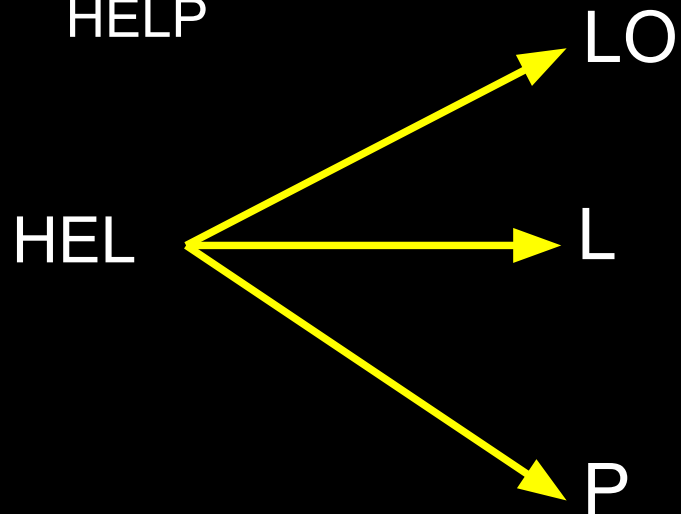


Divergence

How different are two distributions

$$D_{KL}(P||Q) = \sum_i P_i \log \frac{P_i}{Q_i}$$

HELP
HELP
HELLO
HELP



How surprising is this event? $I = -\log_2(p)$

Dataset A



Dataset B



I choose one datapoint randomly then;

1. Probability $p(\text{dog})$
2. Surprise/information/gossipe $-\log (p(\text{dog}))$

Information



$$I_X(x) := -\log[p_X(x)] = \log\left(\frac{1}{p_X(x)}\right)$$

Entropy

$$H(X) := -\sum_{x \in \mathcal{X}} p(x) \log p(x)$$

Mutual Information



$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$$

- X : whether it rains
- Y : whether someone carries an umbrella.

If knowing that someone carries an umbrella makes rain much more likely, then X and Y have high mutual information.

Suppose the true label distribution for a student answer is:

$$P(\text{correct}, \text{wrong}) = [0.8, 0.2]$$

Your model predicts:

$$Q(\text{correct}, \text{wrong}) = [0.6, 0.4]$$

Entropy of the true outcome:

$$H(P) = -0.8 \log_2(0.8) - 0.2 \log_2(0.2) \approx 0.72 \text{ bits}$$

This is the inherent uncertainty: even before seeing the model, the labels are not perfectly certain.

Cross-entropy:

$$H(P, Q) = -0.8 \log_2(0.6) - 0.2 \log_2(0.4) \approx 0.85 \text{ bits}$$

This is the average coding/prediction cost when the model uses Q .

KL divergence:

$$D_{KL}(P\|Q) = H(P, Q) - H(P) \approx 0.85 - 0.72 = 0.13 \text{ bits}$$

Recap

$$y = xW + b$$

$$y_1 = \text{softmax}(y, \text{dim}=1)$$

Layer (macro) view

$$W^1 = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix}, b^1 = [-1.5 \ -0.5 \ 0.5]$$

$$\bar{h}^1 = [\bar{h}_1^1 \ \bar{h}_2^1 \ \bar{h}_3^1]$$

$$= [x_1 \ x_2 \ x_3] \begin{bmatrix} 1 & 1 & -2 \\ -1 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix} + [-1.5 \ -0.5 \ 0.5]$$

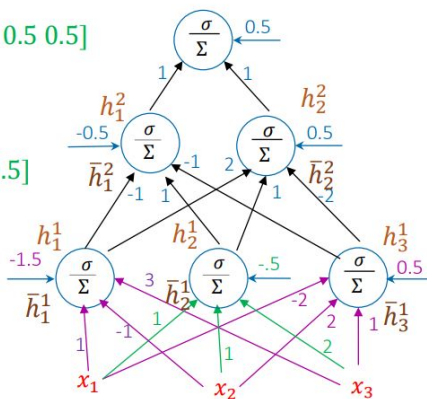
$$= W^1 x + b^1$$

$$h^1 = \sigma(\bar{h}^1)$$

$$W^2 = \begin{bmatrix} -1 & 2 \\ 1 & 1 \\ -1 & -2 \end{bmatrix}, b^2 = [-0.5 \ 0.5]$$

$$\bar{h}^2 = [\bar{h}_1^2 \ \bar{h}_2^2] = h^1 W^2 + b^2$$

Neuron (micro) view



$$x = [x_1 \ x_2 \ x_3]$$

$$W_1^1 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}, b_1^1 = -1.5$$

$$\bar{h}_1^1 = x W_1^1 + b_1^1$$

$$h_1^1 = \sigma(\bar{h}_1^1)$$

$$W_2^1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, b_2^1 = -0.5$$

$$\bar{h}_2^1 = x W_2^1 + b_2^1$$

$$h_2^1 = \sigma(\bar{h}_2^1)$$

$$W_3^1 = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}, b_3^1 = 0.5$$

$$\bar{h}_3^1 = x W_3^1 + b_3^1$$

$$h_3^1 = \sigma(\bar{h}_3^1)$$

1. Linearity
2. Activation Function
3. Batch Normalization

Standardization

Batch-Normalization

$$X = \begin{bmatrix} 1 & 100 \\ 2 & 200 \\ 3 & 300 \\ 4 & 400 \end{bmatrix}$$

$$X_{\text{std}} = \begin{bmatrix} \frac{1-2.5}{1.118} & \frac{100-250}{111.8} \\ \frac{2-2.5}{1.118} & \frac{200-250}{111.8} \\ \frac{3-2.5}{1.118} & \frac{300-250}{111.8} \\ \frac{4-2.5}{1.118} & \frac{400-250}{111.8} \end{bmatrix}$$

$$X_{\text{std}} = \begin{bmatrix} -1.34 & -1.34 \\ -0.45 & -0.45 \\ 0.45 & 0.45 \\ 1.34 & 1.34 \end{bmatrix}$$

Input $\rightarrow$ Linear $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Linear

$$y = \gamma \hat{x} + \beta$$

Okay so far?

What if x , i.e. dataset, is 10M_by_2B matrix?

```
y = xW + b  
y_1 = softmax(y, dim=1)
```

$x_{2B \times 10M}$

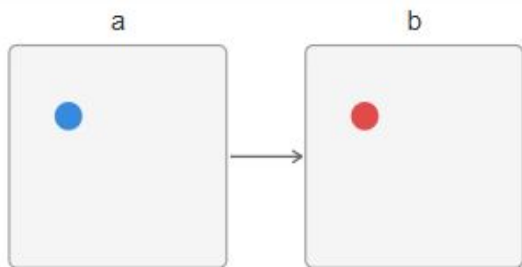
What's the number of
parameters?

Huge, right?

Dataset Operations

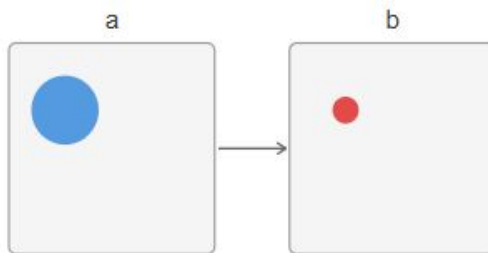
Point

Output value at (m_0, n_0) *only* depends on the input value at the same coordinate.



Local

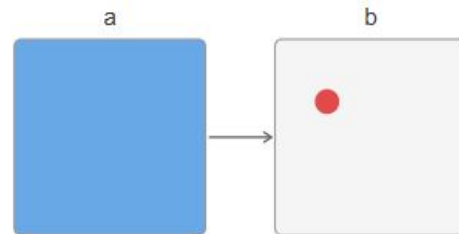
Output value at (m_0, n_0) depends on the input value at the $p \times p$ neighborhood.



Translation Equivariance: pixels close to each other are fundamentally similar

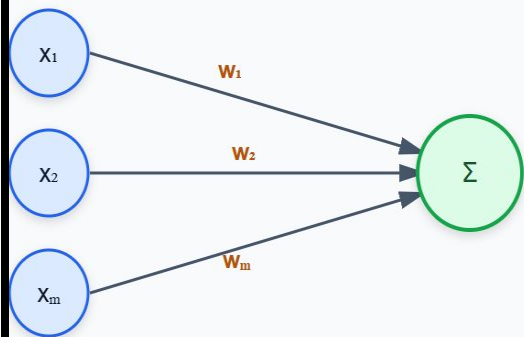
Global

Output value at (m_0, n_0) depends on the $N \times N$ image.



Perceptron: every input has its own weight

Fully connected input $\rightarrow$ output



$$z = w_1x_1 + w_2x_2 + \dots + w_mx_m + b$$

Number of weights grows with input size

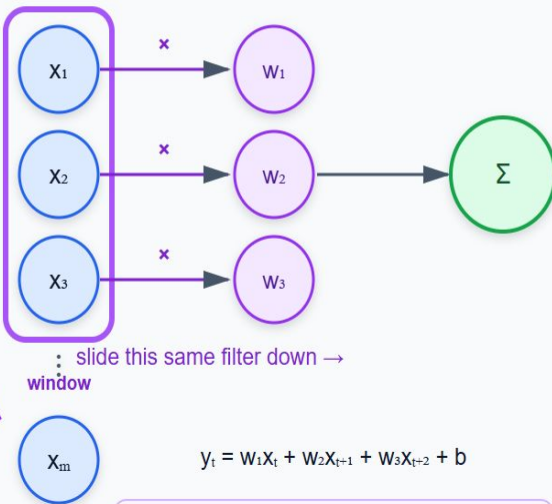
m
Feature
values

and

m
weights

Convolution: a small kernel slides across inputs

A three-weight filter connects to a local input window



$$y_t = w_1x_t + w_2x_{t+1} + w_3x_{t+2} + b$$

Only 3 weights, regardless of sequence length

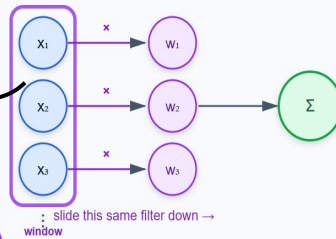
m
Feature
values

but

3
weights

Convolution: a small kernel slides across inputs

A three-weight filter connects to a local input window



: slide this same filter down →
window

$$y_i = w_1x_i + w_2x_{i-1} + w_3x_{i-2} + b$$

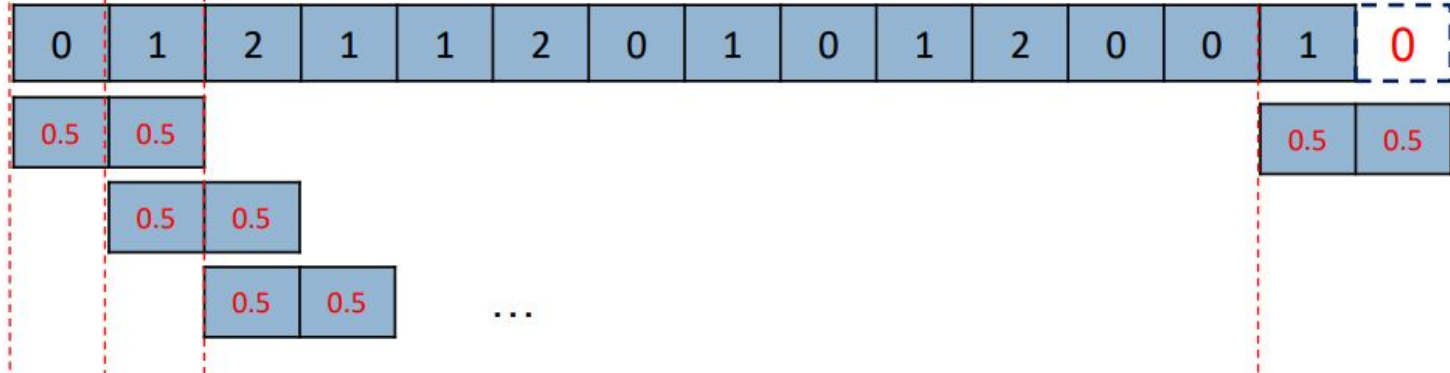
Only 3 weights, regardless of sequence length

What to do when it is not perfect fit?

size = 1x14

filter = 1x2

stride = 1



Human binocular vision pipeline

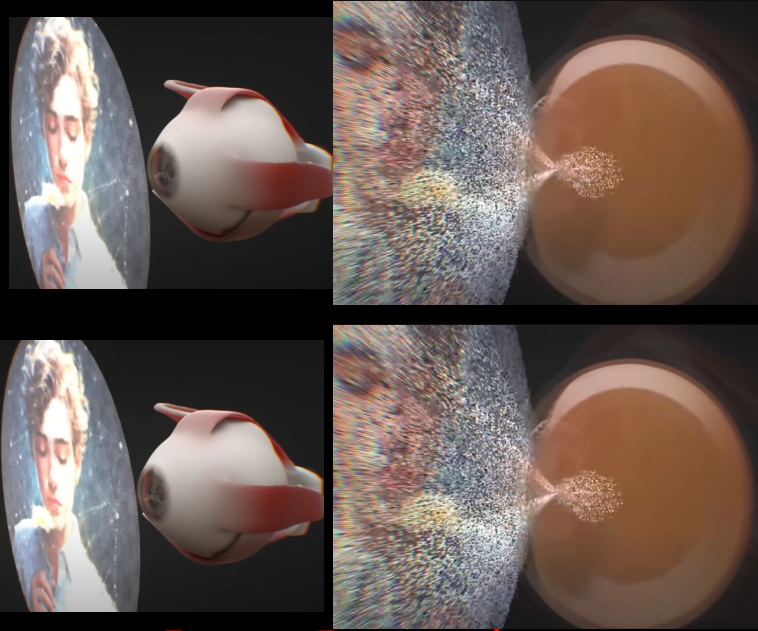
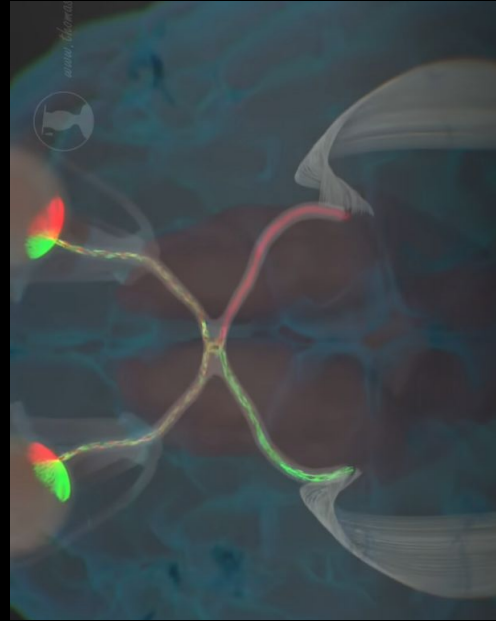


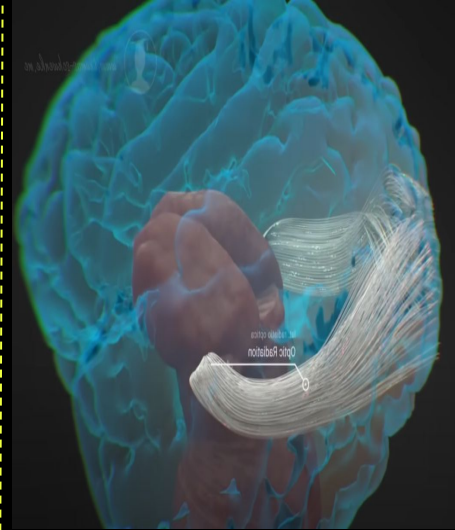
Image Processing
Zone

*how to process the
pixels.*



Computer Vision
Zone

*Extract meaning from
visual data.*



Deep Learning
Zone

Decision

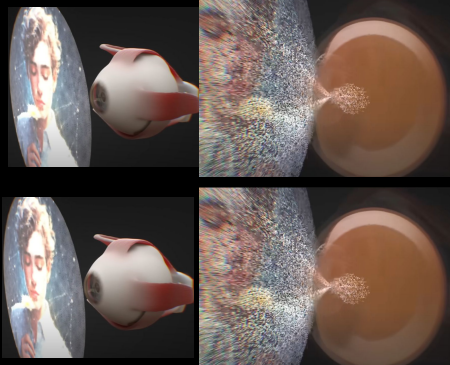
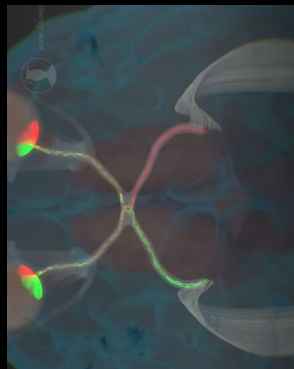
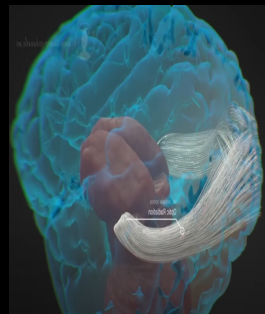


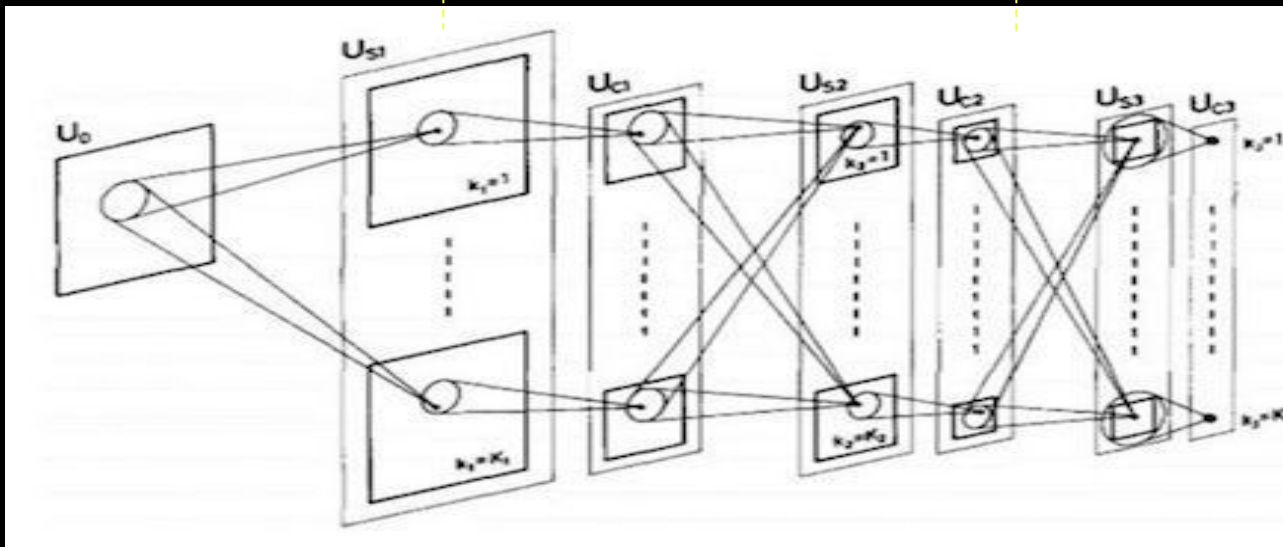
Image Processing
Zone



Computer Vision
Zone



Deep Learning
Zone



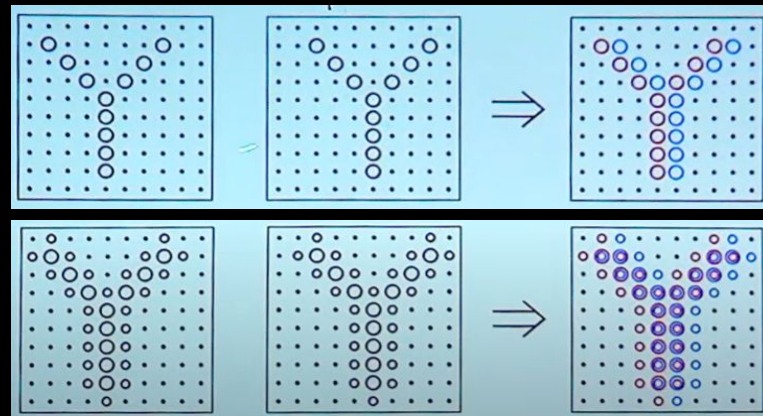
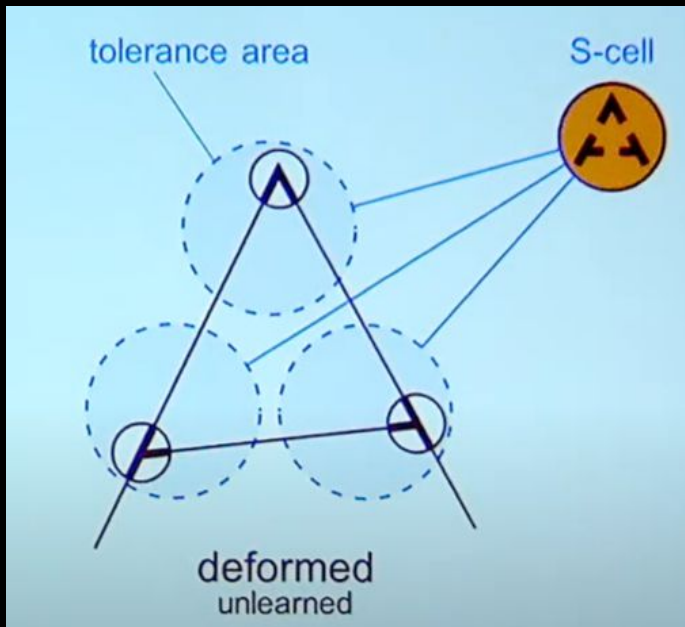
Neocognitron,
1982
Fukushima

Our Eye Has Two Cells

1. Simple Cell (S): Extract important information, features
2. Complex Cell (C): Shift and size tolerance

Our Eye Complex Cell

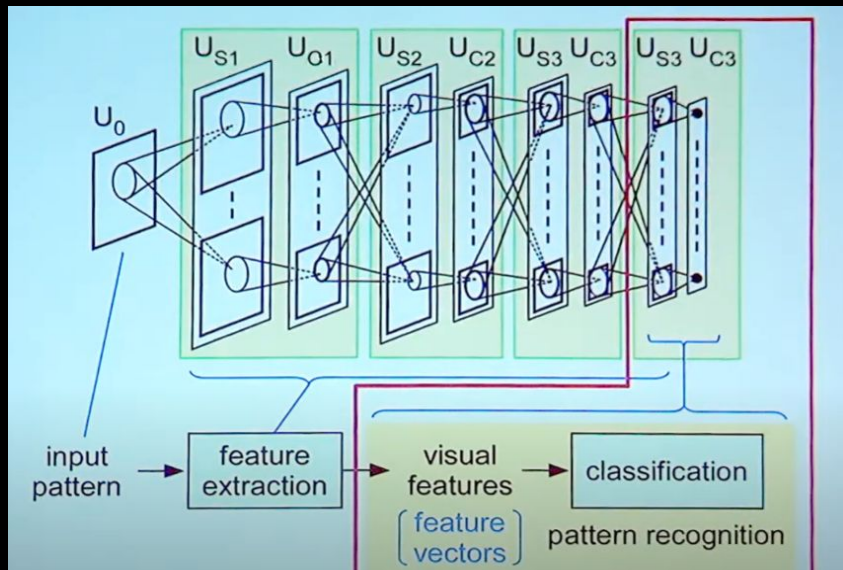
A. Shift Invariance



B. Size Invariance



Neocognitron



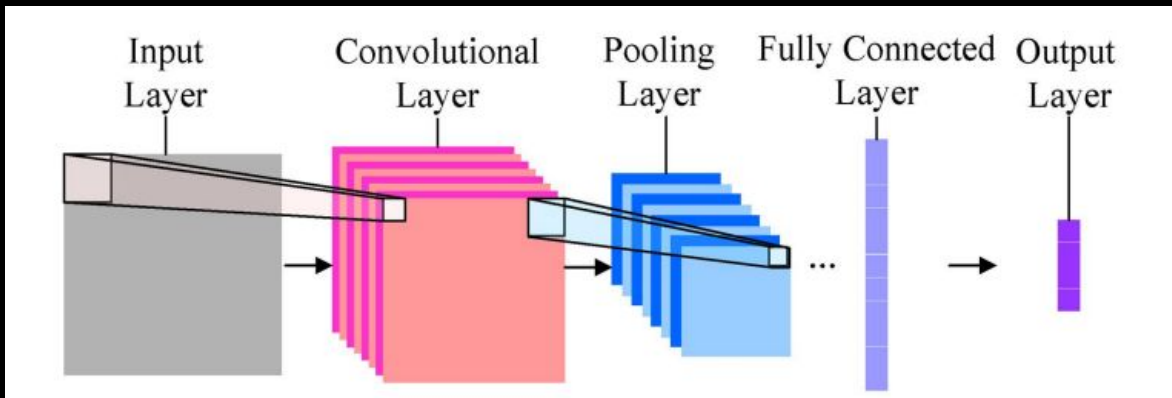
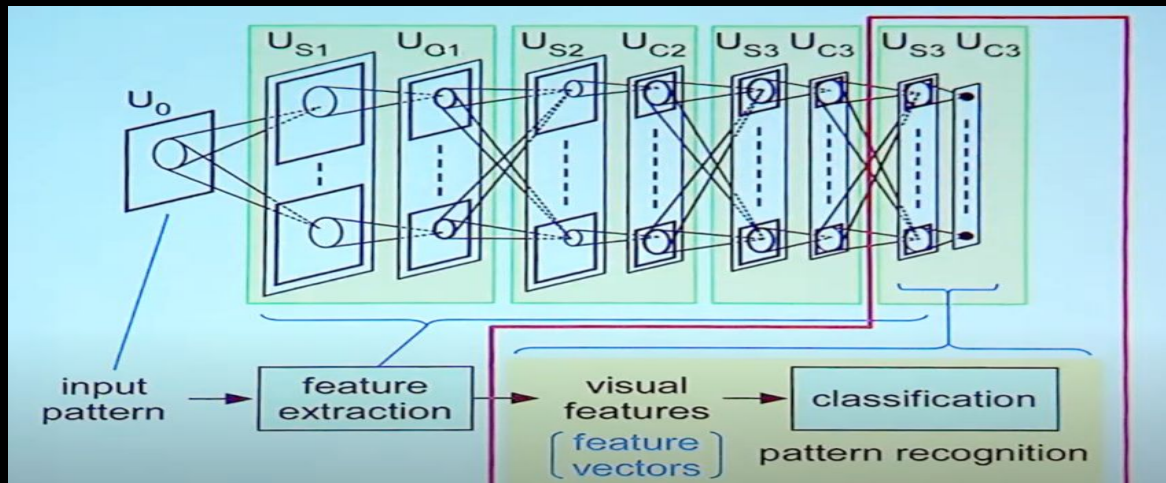
No supervised end-to-end learning

No gradient-based optimization

Limited invariance generalization

Data limitations

From Neocognitron to Convolutional Neural Network



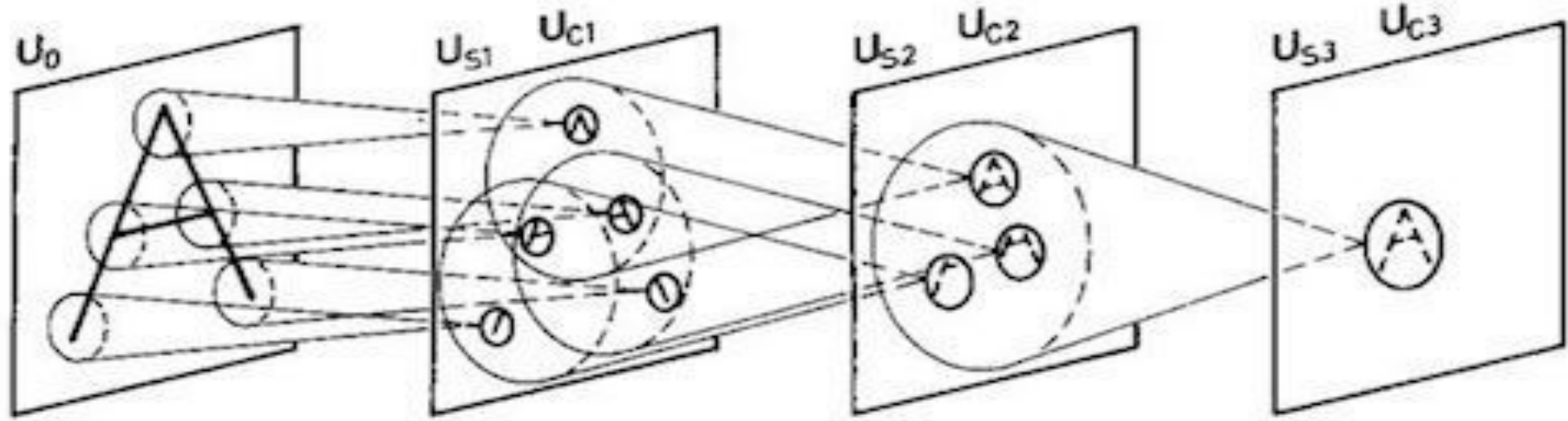


Image Processing
Zone

Computer Vision
Zone

Deep Learning
Zone

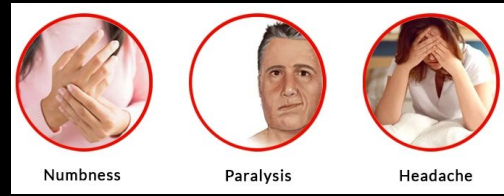
*how to process the
pixels.*

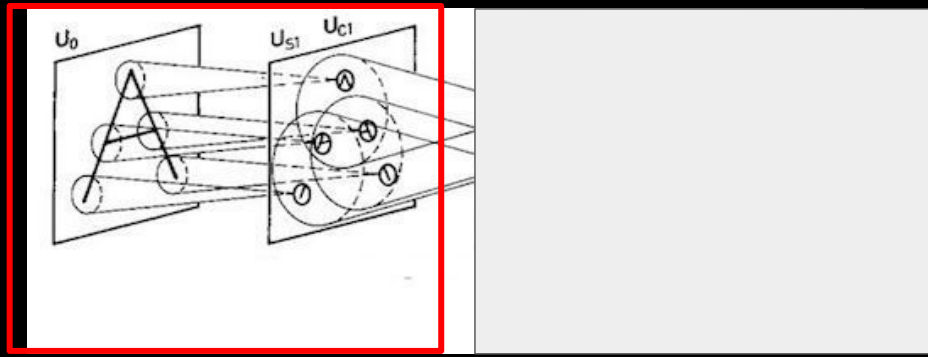


*Extract meaning from
visual data.*



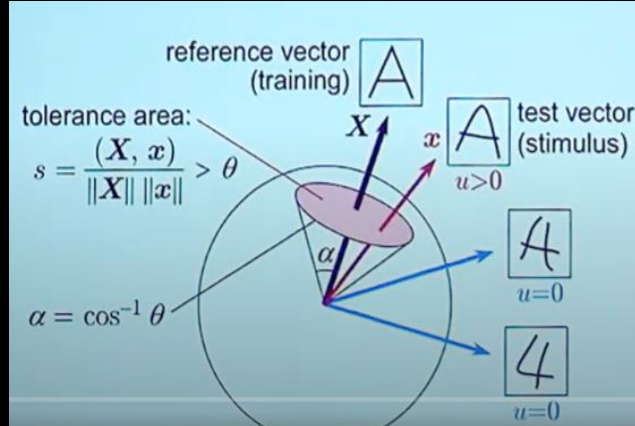
Decision



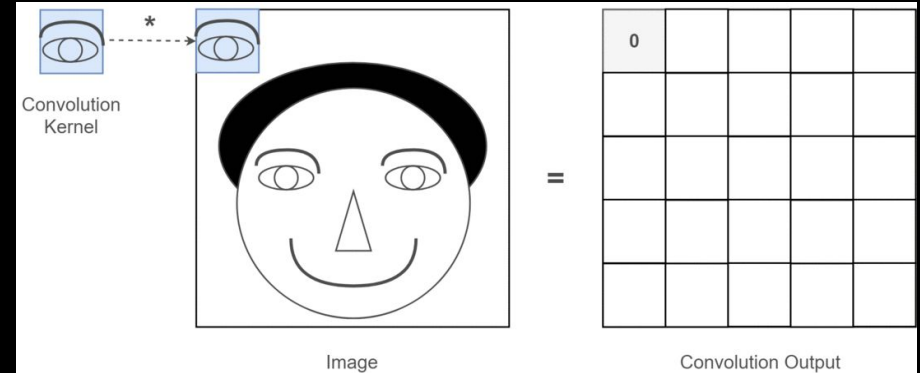


Complex cell

Tolerance



Convolutional + Pooling

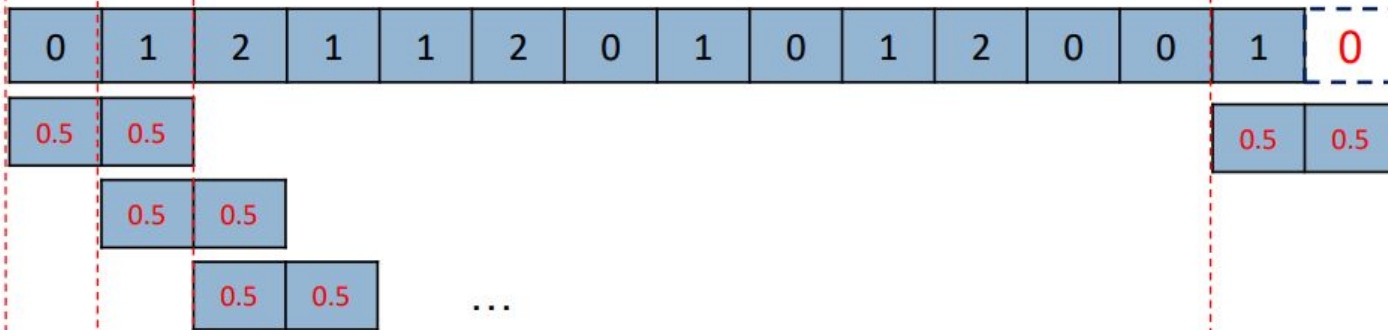


What to do when it is not perfect fit?

size = 1x14

filter = 1x2

stride = 1



Is convolution shift invariant?

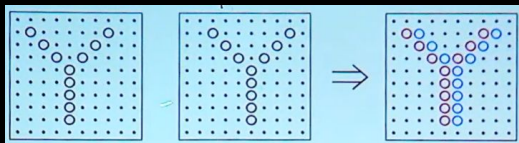
Complex Cell

A. Shift $\rightarrow$ Convolution

$[0, 0, 1, 2, 3, 4, 0]$

$[-1, 0, 1]$

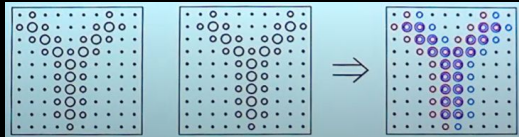
$[1, 2, 2, 2, -3]$



$[0, 0, 0, 1, 2, 3, 4]$

$[-1, 0, 1]$

$[0, 1, 2, 2, 2, -3]$



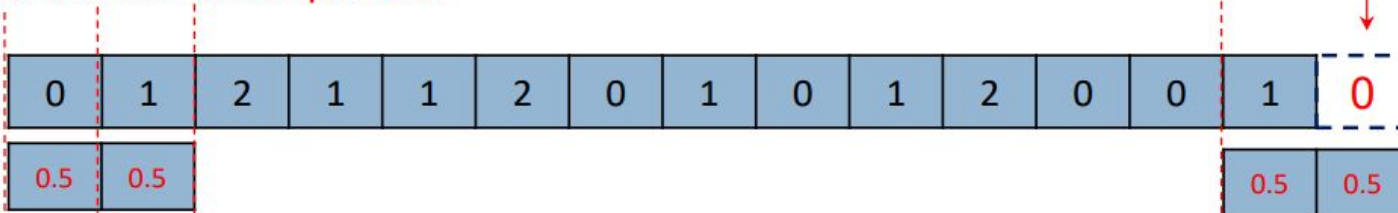
B. Size $\rightarrow$ Pooling



What to do when it is not perfect fit?

size = 1x14

filter = 1x2



Convolution layer with zero padding

x (input tensor (7,7)) strides = (2,2)

0	0	0	0	0	0	0	0	0
0	1	-2	-1	5	3	2	1	0
0	1	3	-1	4	3	3	1	0
0	1	-2	1	6	3	3	2	0
0	2	-2	2	5	2	1	0	0
0	0	3	-2	5	4	1	2	0
0	1	2	-3	1	1	2	-1	0
0	1	-2	-1	1	2	1	1	0
0	0	0	0	0	0	0	0	0

$W_i = 7$

$p = 1$ zero padding

$H_i = 7$

kernel size = (3,3)
 W (filter or kernel)

1	1	1
1	1	1
1	1	1

$f_h = 3$

$f_w = 3$

feature map

3	8	19	7
3	16	30	10
6	11	22	5
2	-2	8	3

$H_o = 4$

$W_o = 4$

- The **sliding window** moves from **left to right, top to bottom** with strides.
- We **convolve** the **filter** and the **sliding windows** to work out the **neurons** on the **feature map**.

